# Routing Algorithms for a HyperX Network

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## 1 Introduction

This project implements a 256-endpoint three-dimensional HyperX network in Garnet [6] and compares four routing algorithms: DOR, Valiant, DimWAR, and OmniWAR. The same platform is also used to study two extensions of OmniWAR: Two-Hop Lookahead and Elastic VC Allocation.

The highlights of the project are:

- a HyperX topology in Garnet;
- four routing implementations with explicit deadlock-avoidance rules;
- two OmniWAR extensions, one broadening the legal VC pool and one adding one-hop lookahead;
- a common comparison using throughput, latency, path length, and detour measurements.

## 2 Architecture

### 2.1 Topology: HyperX

We use the regular HyperX family $H(L, S, T)$ [1]. $L$ is the number of dimensions, $S$ is the number of coordinates in each dimension, and $T$ is the number of terminals attached to each router. Routers are indexed by vectors in $\{0, \ldots, S-1\}^L$. Two routers are directly connected when they differ in exactly one coordinate. Thus, each dimension is a complete graph rather than a line of nearest-neighbor links.

In this project, we use

$$L = 3, \qquad S = 4, \qquad T = 4,$$

so it is a $4 \times 4 \times 4$ HyperX with 64 routers and 256 endpoints. The shortest router path changes each mismatching coordinate once, and therefore has at most three hops. Compared with mesh or torus networks of the same coordinate size, HyperX uses a higher radix and more links in exchange for shorter paths. Mesh and torus are not experimental baselines in this project.

### 2.2 Routing Algorithms

#### DOR

DOR follows a fixed X-Y-Z dimension order, as in conventional dimension-order routing [3]. At each router it corrects the first coordinate that is still different from the destination. Because a HyperX dimension is fully connected, one correction takes one hop. The path is minimal and has a maximum of three hops.

**Statement.** *DOR is deadlock-free under the X-Y-Z channel order.*

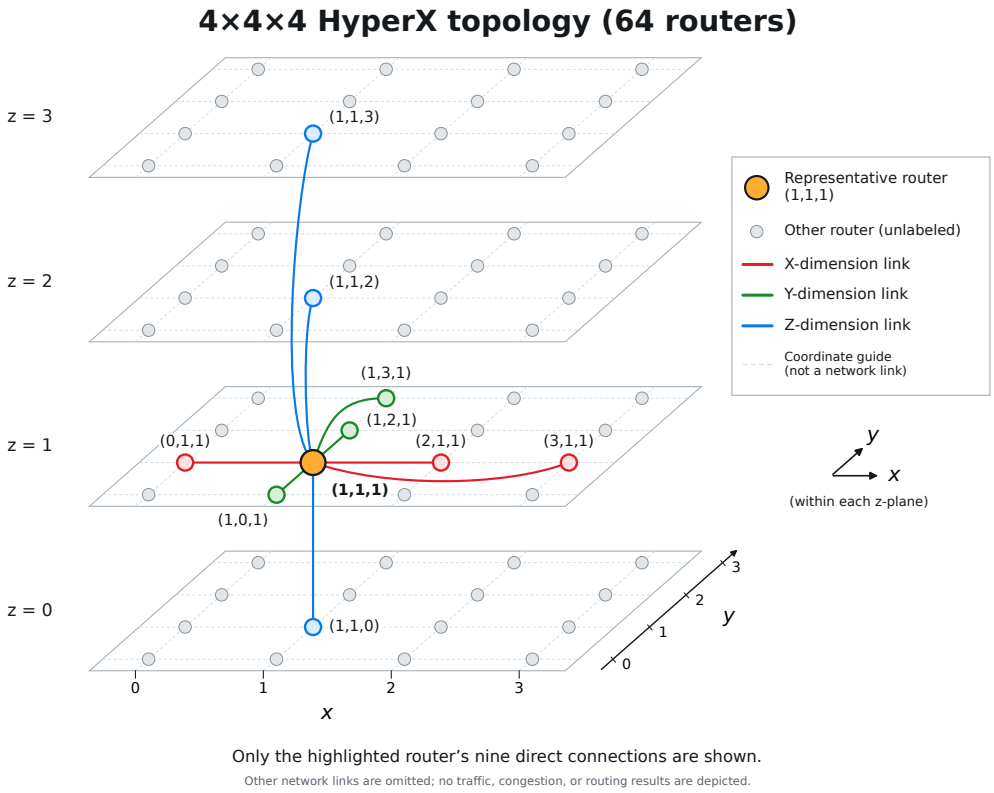



Figure 1: The evaluated HyperX topology.

*Proof.* A packet can request channels in nondecreasing dimension order only. A channel dependency therefore goes from X to Y, from Y to Z, or remains within one dimension; it cannot return to an earlier dimension. The channel-dependency graph is acyclic, so a cyclic wait cannot occur [2]. □

### Valiant

Valiant first selects an intermediate router and then routes from the source to the intermediate and from the intermediate to the destination using DOR [4]. The eight physical VCs are divided into two phases: phase 0 uses VC 0–3 and phase 1 uses VC 4–7. A packet can move from phase 0 to phase 1, but not in the opposite direction. The two DOR legs give a six-hop upper bound.

**Statement.** *The two-phase Valiant implementation is deadlock-free.*

*Proof.* Within each phase, requests follow the acyclic X-Y-Z order. The only cross-phase dependency is phase 0 to phase 1; phase 1 never requests phase 0. A cycle would have to contain a phase-decreasing edge, which is forbidden. Thus the combined dependency graph is acyclic. □

### DimWAR

DimWAR [5] considers the first unaligned dimension. Its candidates are the productive hop in that dimension and the legal detours in the same dimension. For a candidate output and a legal local VC set $M$, the implementation defines

$$Q = \min_{v \in M} \left( 1 - \frac{\text{available credits}(v)}{\text{VC capacity}(v)} \right).$$

The distance term $H$ includes the hop about to be taken: for a productive candidate, $H$ is the shortest distance from the current position to the destination; for a detour candidate, $H$ is this distance plus one. For example, with two unaligned coordinates remaining, a productive candidate has $H = 2$ and a detour candidate has $H = 3$. For each candidate it computes

$$QH = Q \cdot H,$$

then uses the minimum $QH$, the minimum remaining distance $H$, and finally a random tie-break. A dimension can be detoured at most once, giving a maximum path length of six hops.

**Statement.** *DimWAR is deadlock-free when its productive, detour, and repair states advance monotonically.*

*Proof.* Productive requests advance the current dimension toward the destination. A detour enters a separate detour class, and the following repair returns to productive progress without allowing the same dimension to be detoured again. The state order therefore has no transition back to an earlier resource class. Since every channel dependency follows this order, the dependency graph has no directed cycle. □

#### OmniWAR

OmniWAR considers productive and detour candidates in every currently unaligned dimension. Candidates are ranked with the same $QH$, $H$, and final tie-break rule as DimWAR. In the paper-style mapping used by our implementation, six logical classes are mapped to the physical VCs [5]:

$$C_0 \to \{VC_0, VC_6\}, \quad C_1 \to \{VC_1, VC_7\},$$

$$C_i \to VC_i, \qquad i = 2, 3, 4, 5.$$

The logical class advances at each internal hop, and the maximum path length is six hops.

**Statement.** *OmniWAR is deadlock-free because every internal hop advances the logical VC class.*

*Proof.* A packet requesting a channel in class $i$ can next request only a class greater than $i$. The class index is bounded by five, so a dependency path is strictly increasing and finite. No dependency can return to a lower class; hence the channel-dependency graph is acyclic. □

## 3 Improving OmniWAR

### 3.1 Elastic VC Allocation

In paper-style OmniWAR, each logical class is bound to a fixed subset of VCs. A packet may therefore be unable to use an idle VC assigned to another class, even when using that VC would still leave enough classes to finish the route. Elastic VC Allocation removes this fixed binding. At each hop, it exposes an interval of eligible VCs, leaving room for the remaining minimal route under a fixed three-class deroute reserve. The motivation is therefore a wider VC choice space and better sharing of buffer resources across routing stages.

Let $v_{\min}$ be the lowest currently allowed physical class and $H_{\text{after}}$ the remaining minimal distance after a candidate hop. The implementation evaluated in this report uses a fixed reserve of three classes:

$$v_{\max} = \max\left(v_{\min}, 7 - H_{\text{after}} - 3\right),$$

and permits every output VC in the interval

$$v_{\text{out}} \in [v_{\min}, v_{\max}].$$

The lower bound is one greater than the previously allocated physical VC class after the first hop. The subtraction of 3 is fixed and does not decrease when a deroute is consumed. The maximum with $v_{\min}$ retains the lowest eligible class; separate checks ensure enough classes remain to finish minimally. Productive and legal detour candidates are otherwise generated across the unaligned dimensions.

The implementation separates VC eligibility from candidate selection. The interval determines which physical VCs may be requested, while the OmniWAR score selects the output and candidate type. Compared with the fixed paper-style mapping, this interval can expose several feasible VCs to the same routing stage. It expands the resources visible to the congestion score without changing the score definition or the set of candidate output directions.

**Statement.** *The current Elastic VC rule is deadlock-free under the implemented interval.*

*Proof.* Each subsequent physical VC is at least $v_{\min}$, which is one greater than the previously used VC. Thus, a packet's VC number strictly increases at every internal hop. Route checks retain enough classes to finish the remaining minimal path. A dependency cycle would require a decrease in physical VC number, which the rule forbids. □

### 3.2 Two-Hop Lookahead

The motivation for Two-Hop Lookahead is that a first hop can look attractive locally while leading immediately to a busy second hop. The original OmniWAR score uses only the congestion observed for the current candidate. Lookahead keeps the same legal actions and VC ordering, but adds one step of information when ranking the first hop. For a legal first-hop candidate $c_1$, it examines the legal second-hop (output, VC) candidates at the next router and uses their minimum QH:

$$QH_2^{\min}(c_1) = \min_{c_2 \in C_2(c_1)} QH(c_2).$$

The first-hop score is

$$\text{Score}(c_1) = QH(c_1) + \gamma QH_2^{\min}(c_1).$$

Full L2 applies this query to all first-hop candidates. Finalist L2 applies it only to the candidates that remain after the original OmniWAR ranking.

Because the purpose is to evaluate additional information rather than additional resources, the lookahead query is read-only: it must not consume credits, change allocator state, or commit the packet's route state. Full L2 maximizes candidate coverage but performs more remote reads; Finalist L2 tests whether most of the effect can be retained with a smaller query set.

## 4 Implementation

### 4.1 Routing Pipeline

At each route computation, the routing unit determines the next router, output direction, legal VC set, and packet state. The output direction identifies a dimension and a peer coordinate, so a selected HyperX candidate maps to one concrete Garnet output port. A stalled request keeps its selected candidate; it is not re-sampled on every allocator retry. State is committed only after the hop actually arrives.

### 4.2 VC and State Management

Valiant stores its intermediate router and phase. DimWAR stores the current dimension and whether a detour has already been used there. OmniWAR stores its logical class. Elastic VC Allocation stores the current physical class and constructs the feasible VC interval at each hop. Output VC allocation is restricted to the set returned by routing.

### 4.3 Congestion-Based Selection

For a legal physical $(\text{output}, VC)$ pair, the congestion value is

$$Q = 1 - \frac{\text{available credits}}{\text{VC capacity}}.$$

The distance term $H$ includes the candidate hop: it is one plus the minimal distance from the next router to the destination. All legal VCs on an output have the same $H$, so the implementation uses their minimum per-VC $Q$. DimWAR and OmniWAR use this value for the first-hop $QH$ score. Two-Hop Lookahead repeats this operation for second-hop candidates.

### 4.4 Deadlock Enforcement

The implementation follows a monotonic resource order for each routing family:

- DOR advances in X-Y-Z order;
- Valiant advances from phase 0 to phase 1;
- DimWAR advances through its dimension and detour/repair state;
- OmniWAR advances its logical class;
- Elastic VC Allocation strictly increases the physical VC class.

Runtime checks enforce legal output directions, legal VC sets, state transitions, and the maximum hop bound.

## 5 Evaluation

### 5.1 Experimental Setup

All formal comparisons use the same HyperX configuration, four buffer slots per control VC, one-flit packets, 16-byte links/flits, and unit link/router latency. Each offered rate uses a warm-up interval followed by two consecutive measurement windows. The tested rates run from 0.01 to 0.60.

The main workload comparison uses Bit Complement, Dimension Complement Reverse (DCR), and Transpose.

### 5.2 Traffic Patterns

The traffic generator uses fixed endpoint permutations. For an endpoint ID, the low two bits identify the terminal and the remaining bits identify the router coordinates $(x, y, z)$.

- **Bit Complement**: all endpoint bits are complemented, $s \mapsto s \text{ xor } 255$.

- **DCR**: the terminal is preserved and the three router coordinates are complemented and reversed, $(x, y, z) \mapsto (3 - z, 3 - y, 3 - x)$.
- **Transpose**: the low and high four-bit halves of the complete endpoint ID are exchanged:

$$d = ((s \,\&\, \texttt{0x0f}) \ll 4) \mid ((s \,\&\, \texttt{0xf0}) \gg 4).$$

  With the two-bit fields ordered as $[z, y, x, t]$ from most to least significant, this is equivalent to $(t, x, y, z) \mapsto (y, z, t, x)$. The terminal field is therefore not preserved.

Each mapping is a bijection, so every endpoint has exactly one destination.

### 5.3 Main Metrics

The main metrics are accepted throughput (reception rate), average packet latency, average router hops, and deroute fraction. Delivery ratio is used to check that a run completed normally. Candidate counts and remote Q reads are reported only when explaining the cost of an OmniWAR extension.

### 5.4 Valid and Stable Points

Valid means that a run produced parseable data. Stable additionally means that the two consecutive measurement windows show no continuing growth in throughput or latency. Throughput plots show all valid points and distinguish stable from unstable points. Latency, hops, and deroute plots use stable points in the main text; all-valid diagnostic versions are supplementary because post-saturation windows contain growing queues rather than a steady-state measurement.

### 5.5 Baseline Routing Results

Figure 2 shows the complete five-routing throughput comparison, and Figures 3 and 4 show the stable latency and hop-count comparisons. The highest tested stable offered rates are:

Table 1: Highest tested stable offered rate.

| Traffic | DOR | Valiant | DimWAR | OmniWAR | Elastic |
|---|---|---|---|---|---|
| Bit Complement | 0.20 | 0.25 | 0.30 | 0.25 | 0.25 |
| DCR | 0.06 | 0.25 | 0.08 | 0.20 | 0.25 |
| Transpose | 0.15 | 0.30 | 0.10 | 0.35 | 0.40 |

DOR has the lowest latency at low load because it adds no intermediate router or adaptive detour. At higher load, the best stable offered rate depends on the traffic. Among the four original routing algorithms, DimWAR is best for Bit Complement, Valiant is best for DCR, and OmniWAR is best for Transpose. No original routing algorithm wins on all three patterns.

The different winners are more informative than a single overall ranking. DOR pays almost no routing overhead when the network is lightly loaded, so its short minimal paths are visible directly in the latency curves. As injection increases, however, the traffic permutation determines which links become the limiting resources. The results are consistent with DimWAR being effective for Bit Complement and with Valiant's path diversity helping DCR. Transpose is the case where OmniWAR's larger candidate set is useful: it can choose among unaligned dimensions instead of being tied to the first one.

The hop-count curves help interpret these results. We interpret a throughput difference together with average hops and deroute fraction, rather than treating throughput alone as evidence of better routing. Conversely, the longer paths of Valiant are a deliberate cost of load spreading. The experiments therefore support a traffic-dependent routing choice, not a universally best algorithm.

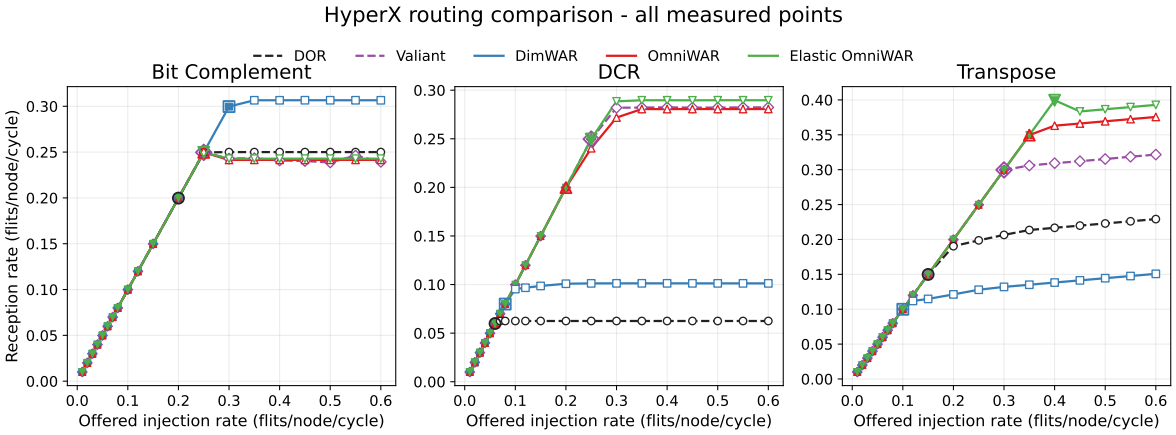



Figure 2: Throughput versus offered injection rate. Filled markers are stable points, open markers are unstable points, and the larger matching-color outline marks the last stable point.

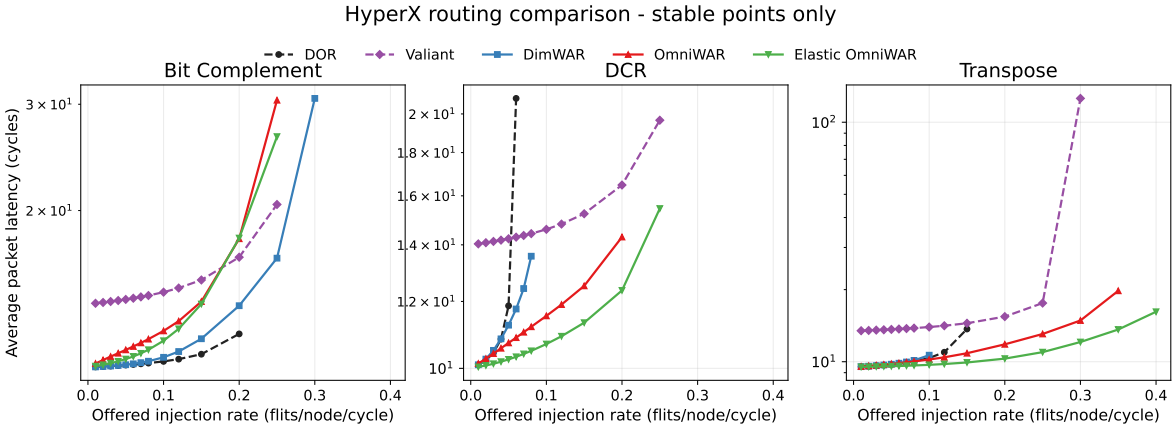



Figure 3: Packet latency on stable points.

## 5.6 Elastic VC Allocation Results

Elastic VC Allocation improves the stable boundary for DCR and Transpose, but not for Bit Complement. Its highest tested stable rates are 0.25 for DCR and 0.40 for Transpose, compared with 0.20 and 0.35 for paper-style OmniWAR. Bit Complement remains at 0.25 for both.

The improvement is consistent with the larger legal VC pool reducing restrictions created by the fixed paper-style class mapping. Because $Q$ is computed over the VCs that a candidate may actually request, exposing more feasible VCs can give a packet access to an idle buffer that would otherwise belong to another logical class. It can also change the QH comparison and therefore the selected path.

The DCR and Transpose results suggest that this additional VC flexibility can relieve some

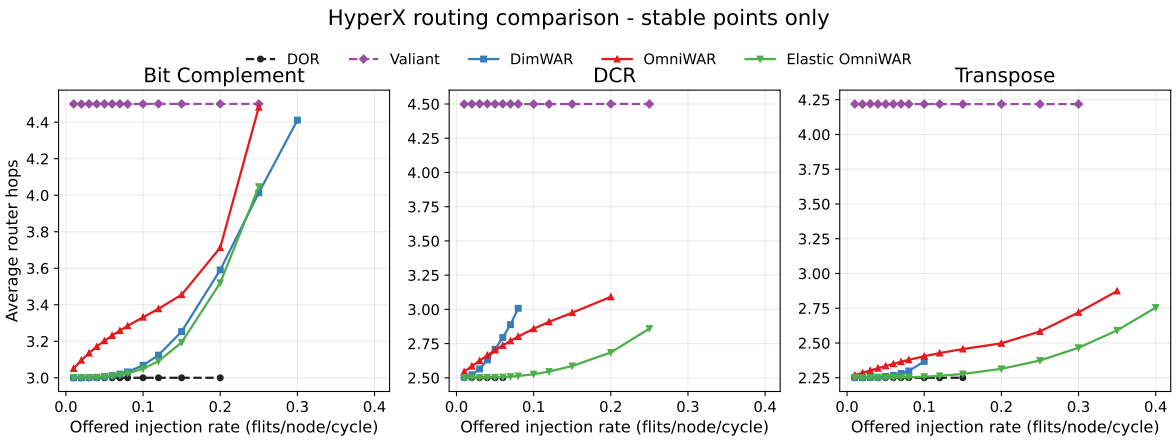



Figure 4: Average router hops on stable points.

class-local buffer pressure. The unchanged Bit Complement boundary shows that a wider VC pool is not sufficient to improve every workload. Elastic VC Allocation does not add physical link bandwidth, and it does not introduce new output directions; it changes which buffers are eligible and, through the QH score, how the existing paths are selected. Its benefit therefore depends on whether fixed VC classes are part of the active bottleneck. At their highest stable points, Elastic reaches average hops of 2.8595 for DCR and 2.7545 for Transpose, with deroute fractions of 0.1257 and 0.1831, respectively.

## 5.7 Two-Hop Lookahead Results

### 5.7.1 Experimental Setting

We design a traffic pattern called `class_aligned_hidden_turn` to study whether information from the next router can improve routing decisions. Each source has a fixed destination, and every endpoint appears exactly once as a destination. Source and destination retain the same terminal index.

The network is a $4 \times 4 \times 4$ HyperX, with four terminals numbered 0–3 attached to each router. For description, routers with $x = 0, 1$ form the cold region, while routers with $x = 2, 3$ form the hot region. The traffic consists of four groups, each containing 64 sources, as shown in Table 2.

Table 2: Traffic groups and example source–destination pairs.

| Traffic group | Sources | Example source → destination |
|---|---|---|
| Cold one-hop background | Cold-region terminals 0 and 1 | `000/t0` → `020/t0`<br>`000/t1` → `030/t1` |
| Hot two-hop background | Hot-region terminals 0 and 1 | `203/t0` → `220/t0`<br>`202/t1` → `230/t1` |
| Cold two-hop targets | Cold-region terminals 2 and 3 | `000/t2` → `210/t2` |
| Hot one-hop traffic | Hot-region terminals 2 and 3 | `200/t2` → `000/t2` |

Here, a coordinate string such as `000` denotes router $(0, 0, 0)$, and `t` identifies the terminal. One-hop and two-hop refer to the shortest router-path distance.

For example, the target flow from 000 to 210 has two minimal routes:

- X first: 000 → 200 → 210;
- Y first: 000 → 010 → 210.

The hot-region background flows can follow 203 → 200 → 220 and 202 → 200 → 230, using class C1 for their second hops, as illustrated in Figure 5. At router 200, these flows share the same class-specific resources as the target flow's legal detour candidates, 200→220 and 200→230. This arrangement aims to give the continuations different loads when the first-hop conditions are similar, providing information that L2 can use when choosing the first hop.

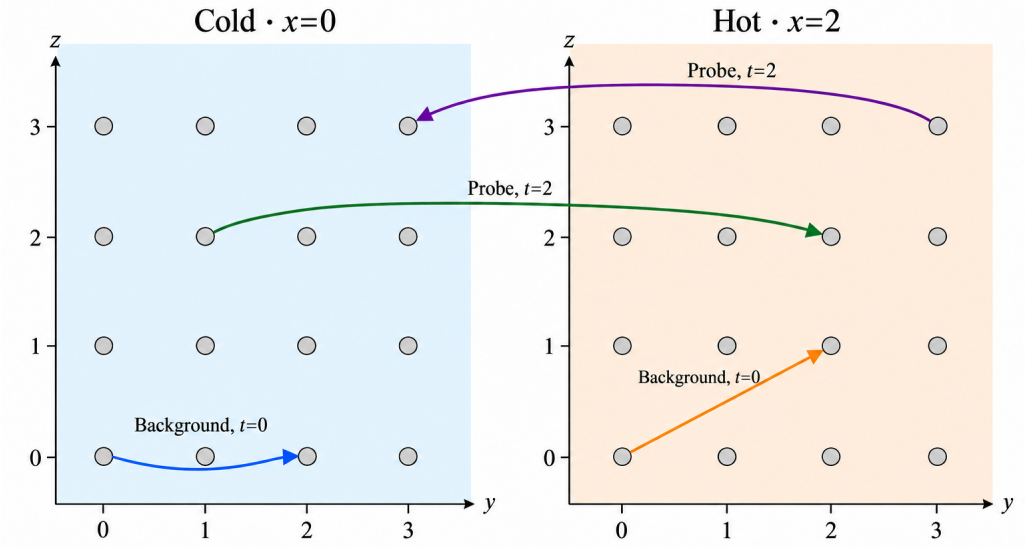



Figure 5: Cold ($x = 0$) and hot ($x = 2$) planes of `class_aligned_hidden_turn`. Arrows show source–destination mappings, not routed paths. Minimal distances: cold background and hot probes, one hop; hot background and cold probes, two hops.

Within each run, both target and background flows use the routing algorithm being evaluated and select their paths according to the network state. We compare OmniWAR, Full L2, and Finalist L2 under the same source–destination mapping.

The experiment uses eight physical VCs, six logical classes, and single-flit packets. Both L2 variants use $\gamma = 1$ throughout each run. The offered injection rate is 0.425, at which all three algorithms were stable in the preliminary screen. Each algorithm is evaluated with seeds 1–3. Every run uses a 50,000-cycle warm-up followed by two consecutive 100,000-cycle measurement windows.

### 5.7.2 Results

At the common point, Full L2 reduces overall average packet latency by about 2.05% and Finalist L2 by about 1.23% relative to OmniWAR. Figure 6 shows the paired latency and path changes. The corresponding average hops are 2.0027 for Full L2, 2.0098 for Finalist L2, and 2.0194 for

OmniWAR. The fractions of internal hops that are deroutes are 0.2510 for Full L2, 0.2536 for Finalist L2, and 0.2572 for OmniWAR. Figure 7 summarizes lookahead work. Finalist L2 uses 16.25% of Full L2's remote $Q$ reads per route computation.

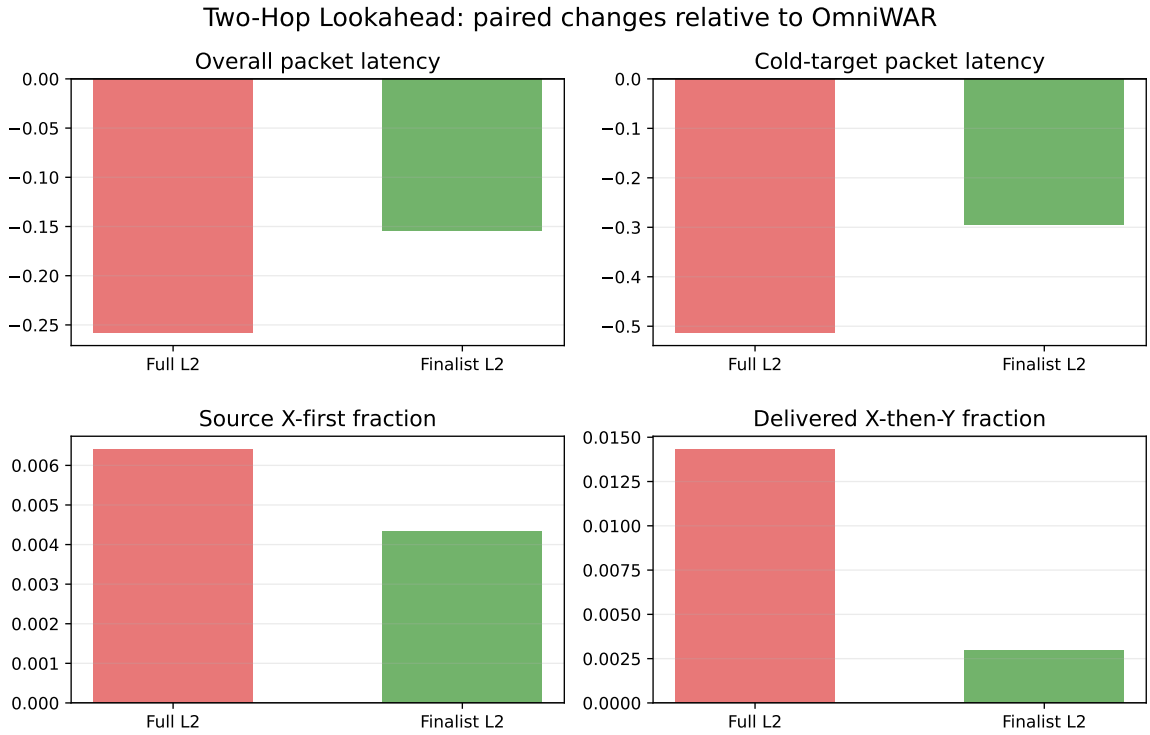



Figure 6: Paired latency and path comparison for the class-aligned traffic.

### 5.7.3 Analysis

Full L2 reduces overall average packet latency from 12.571 to 12.313 cycles, an improvement of approximately 2.05%, while Finalist L2 improves it by 1.23%. These modest gains suggest that additional second-hop information does not translate into a substantial latency advantage in this experiment. Several factors may explain the limited benefit.

First, the intended second-hop congestion contrast is not consistently present. The traffic was designed so that choosing X first would lead toward a busier second hop, while choosing Y first would avoid it. However, background packets also use adaptive routing rather than fixed paths through the intended bottleneck. When comparing the two shortest-path continuations, the second-hop Q after X-first exceeds that after Y-first in only about 32% of observations. Thus, the source–destination mapping does not guarantee a persistent distinction between a congested route and an uncongested alternative, limiting the opportunities for lookahead to help.

Second, actual routing is not a binary choice between X-first and Y-first. OmniWAR already supports derouting: approximately 40.4% of initial decisions for target packets select a deroute, compared with 39.9% under Full L2. Meanwhile, the selection rate of the predefined congested X-first branch increases from approximately 34.3% to 34.9%. The results therefore do not match the expectation that L2 would redirect substantial traffic away from that branch. The overall path-length change is also small: average hops decrease from 2.0194 to 2.0027, a reduction of only about 0.83%.

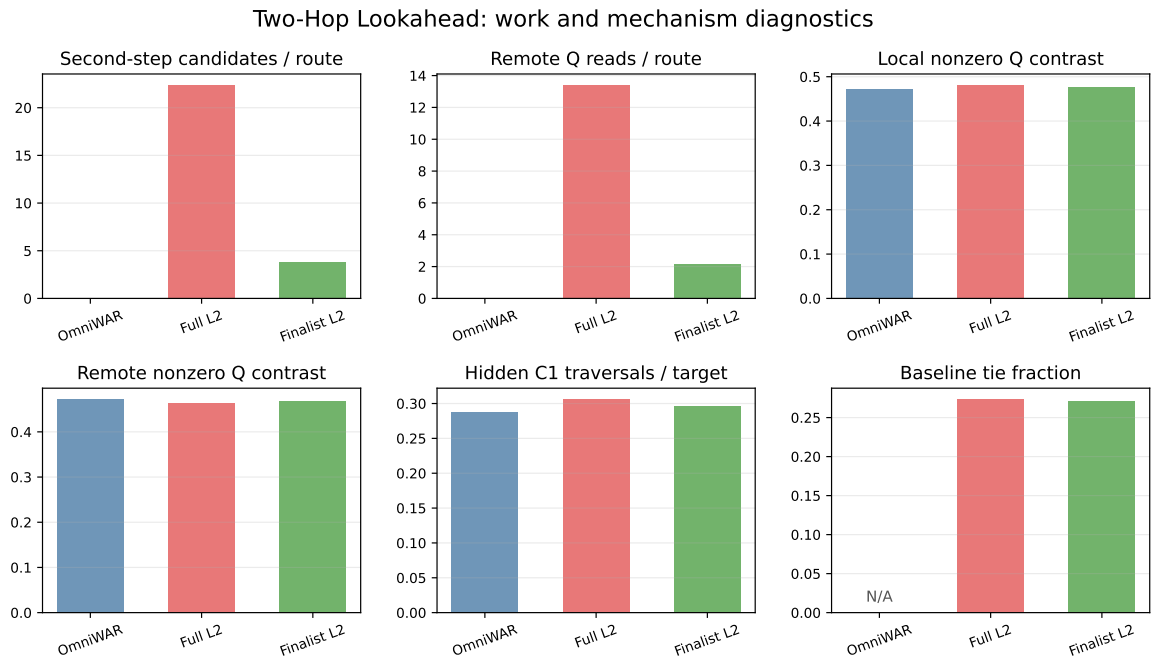



Figure 7: Lookahead work and mechanism diagnostics. N/A indicates that the L2-specific baseline-tie fraction is not measured for OmniWAR; it does not indicate zero ties.

Finally, Q measures current VC occupancy, not the waiting time a packet will experience upon reaching the second hop. An occupied VC may become available before arrival, while a currently available VC may be taken by another packet. Different VCs also share the same physical link, so an available VC does not guarantee immediate transmission. Consequently, choosing a lower-Q continuation need not produce a shorter actual wait. This mismatch may allow some decisions to improve latency while others increase it, limiting the net benefit.

Overall, the small improvement may reflect a combination of an inconsistent congestion contrast, OmniWAR's existing routing flexibility, and the gap between current Q and future waiting time. Finalist L2 achieves a 1.23% improvement using only approximately 16.25% of Full L2's remote Q queries, further indicating that the broader query coverage provides limited additional benefit in this experiment.

## 5.8 Final Comparison

The results show that routing quality depends on the traffic pattern. DOR is the lowest-latency choice at light load. At higher load, DimWAR, Valiant, and OmniWAR each lead on at least one baseline workload, so there is no universal winner.

Elastic VC Allocation broadens the feasible VC pool and improves the stable boundary for DCR and Transpose, but not for Bit Complement. Two-Hop Lookahead reduces latency at the selected class-aligned operating point, with a lower-work Finalist variant preserving part of the effect. The results separate three effects: path freedom changes which links can carry traffic, elastic allocation changes which buffers a hop may use, and lookahead changes how a legal candidate is ranked. These effects should therefore be evaluated against the traffic pattern and the cost of the additional decision logic, rather than summarized as one global ranking. Neither extension should be described as a universal capacity improvement.

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## Division of Labor

- **Si Li**: mainly responsible for background research, the four routing algorithm experiments, and Two-Hop Lookahead.
- **Chengwei Liang**: mainly responsible for Elastic VC Allocation, the report, and the presentation.

The project was carried out in close collaboration, with each member also contributing to the other's parts.